

Mathematical and Computational Methods

Exercise Sheet 3: Complex numbers II: exponential form, De Moivre, roots

This sheet accompanies *Unit 3 (Complex numbers II: exponential form, De Moivre, roots)*. It exercises the exponential form, De Moivre's theorem, the n th roots of a complex number, and the trigonometric identities they generate; nothing beyond Units 1–3 is needed.

How to use this sheet. Problems are graded by difficulty—★ drill (routine fluency), ★★ core (standard, tutorial level) and ★★★ challenge (harder or extension)—and organised in three sections: **warm-up drills** for fluency, **tutorial problems** (the core set for the 2-hour class, with the live items flagged ►) and **further practice** for independent home study.

Solutions. A full worked solution sits after each problem: hidden in the student build, shown in the solutions build. To produce the solutions version, uncomment `\solutionson` in the preamble (or build with `pdflatex "\def\showssolutions{}\input{MCM_Exercises_Unit3}"`) and recompile.

Notation. As in the notes, $i^2 = -1$, $e^{i\theta} = \cos \theta + i \sin \theta$, a nonzero z has exponential form $z = r e^{i\theta}$ with $r = |z|$ and $\theta = \arg z$, and $\bar{z}$ is the complex conjugate. Give exact answers (integers, simple surds or fractions) unless a decimal is asked for.

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1 Warm-up drills

Short routine questions, at least one per skill. Aim for speed and accuracy; a few ask for a one- or two-line “show” derivation.

Exercise 1.1 (★ drill — *exponential form*). Write each number in exponential form $r e^{i\theta}$ with $r > 0$ and $-\pi < \theta \leq \pi$:

$$(a) 1 + i, \quad (b) -3, \quad (c) 2i, \quad (d) -1 - i, \quad (e) \sqrt{3} - i, \quad (f) -2 + 2\sqrt{3}i.$$

Exercise 1.2 (★ drill — *back to Cartesian*). Write each in Cartesian form $a + ib$:

$$(a) 2e^{i\pi/3}, \quad (b) 3e^{i\pi}, \quad (c) \sqrt{2}e^{-i\pi/4}, \quad (d) 4e^{i3\pi/2}, \quad (e) e^{i5\pi/6}.$$

Exercise 1.3 (*★ drill — Euler values*). Evaluate $e^{i\pi/2}$, $e^{i2\pi}$, $e^{-i\pi}$ and $e^{i\pi/3}$, and state Euler's identity.

Exercise 1.4 (*★ drill — rotation*). Let $z = 2e^{i\pi/6}$. Write down $e^{i\pi/2}z$ in exponential form, and describe geometrically the effect of multiplying a complex number by (i) i , (ii) $e^{i\pi}$, (iii) $e^{-i\pi/2}$.

Exercise 1.5 (*★ drill — De Moivre*). Use De Moivre's theorem to simplify $(\cos \theta + i \sin \theta)^4$ and $(\cos \theta + i \sin \theta)^{-2}$.

Exercise 1.6 (*★ drill — integer powers*). Compute, using the exponential form:

$$(a) (1 + i)^4, \quad (b) (\sqrt{3} + i)^3, \quad (c) (1 - i)^6, \quad (d) (2e^{i\pi/5})^{10}.$$

Exercise 1.7 (*★ drill — a few roots*). Find all the (a) square roots of i , and (b) fourth roots of 1 .

Exercise 1.8 (*★ drill — Argand sketch*). Plot the four fourth roots of unity $1, i, -1, -i$ on an Argand diagram, mark the unit circle, and name the regular polygon they form.

Exercise 1.9 (*★ drill — counting roots (FTA)*). State, with the fundamental theorem of algebra in mind, how many roots in $\mathbb{C}$ (counted with multiplicity) each has: (a) a degree-5 polynomial; (b) $z^7 - 3 = 0$; (c) $(z - 1)^2(z^2 + 1) = 0$. True or false: every real cubic has at least one real root?

Exercise 1.10 (*★ drill — double angle by De Moivre*). By expanding $(\cos \theta + i \sin \theta)^2$, derive the identities for $\cos 2\theta$ and $\sin 2\theta$.

Exercise 1.11 (*★ drill — hyperbolic link*). Using $\cos \theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta})$, evaluate $\cos(i)$ in terms of e . (The general identity $\cos(ix) = \cosh x$ is built up in the tutorial.)

Exercise 1.12 (*★ drill — phasors*). (a) Write $5 \cos(\omega t + \frac{\pi}{3})$ and $2 \sin \omega t$ as phasors $P = A e^{i\varphi}$. (b) The phasor $3 e^{-i\pi/2}$ represents which real signal $A \cos(\omega t + \varphi)$?

2 Tutorial problems

The core set, to be worked through in the tutorial. Anchored on the worked examples of the notes. Problems marked ► are the core set for the 2-hour tutorial; the rest are for completion at home.

Exercise 2.1 (► ** core — arithmetic in exponential form). Let $z = \sqrt{3} + i$ and $w = 1 + i$. Write z and w in exponential form, then find zw and z/w in both exponential and Cartesian form.

Exercise 2.2 (** core — rotation in action). (a) Rotate $z = 3 + 4i$ anticlockwise by $\frac{\pi}{2}$ about the origin, and then (separately) clockwise by $\frac{\pi}{2}$. (b) Rotate $2 + 2i$ anticlockwise by $\frac{\pi}{4}$. Give exact Cartesian answers.

Exercise 2.3 (** core — prove De Moivre). Prove by induction that $(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$ for all integers $n \geq 1$. (Pure-maths.)

Exercise 2.4 (► ** core — powers by De Moivre (anchor)). Compute $(1 + i)^8$ and $(\sqrt{3} + i)^6$ using De Moivre's theorem.

Exercise 2.5 (** core — a higher power). Evaluate $(-1 + i)^{10}$, and confirm your answer by squaring $-1 + i$ first.

Exercise 2.6 (► ** core — roots, two shapes (anchor)). Find (a) the cube roots of 8 and (b) the fourth roots of -16 . In each case give exact Cartesian values and describe the polygon they form on the Argand diagram.

Exercise 2.7 (► **core — *sixth roots of unity* (anchor)). Find all sixth roots of unity in Cartesian form, and verify that they sum to zero. Why does this generalise to all $n \geq 2$?

Exercise 2.8 (**core — *solving $w^n = z$*). Solve $z^3 = -27i$, giving the three roots in exponential and Cartesian form.

Exercise 2.9 (► **core — *$\cos 3\theta$ and $\sin 3\theta$* (anchor)). Using De Moivre and the binomial theorem, show that $\cos 3\theta = 4\cos^3 \theta - 3\cos \theta$ and $\sin 3\theta = 3\sin \theta - 4\sin^3 \theta$.

Exercise 2.10 (**core — *power reduction*). Use the inverse Euler formula to write $\sin^4 \theta$ as a sum of cosines of multiples of θ .

Exercise 2.11 (► **core — *trig-hyperbolic link*). (a) Show $\cos(ix) = \cosh x$ and $\sin(ix) = i \sinh x$. (b) Hence evaluate $\cos(i \ln 2)$ exactly.

Exercise 2.12 (► **core — *combining oscillations* (anchor; physics)). (a) Combine $3\cos \omega t + 4\sin \omega t$ into a single sinusoid $R \cos(\omega t - \alpha)$. (b) Combine $2\cos \omega t + 2\cos(\omega t + \frac{\pi}{2})$ likewise.

3 Further practice (home)

A larger bank for independent study, routine to hard, with challenge problems marked ***.

Euler, forms and rotation (Skills 1–2)

Exercise 3.1 (*drill — *mixed conversions*). (a) To exponential form ($-\pi < \theta \leq \pi$): -4 , $3i$, $-2 - 2i$, $-1 + \sqrt{3}i$. (b) To Cartesian form: $5e^{i\pi/2}$, $2e^{-i2\pi/3}$, $\sqrt{2}e^{i3\pi/4}$.

Exercise 3.2 (*drill — *index laws*). Simplify to a single $re^{i\theta}$ (then to Cartesian if clean): (a) $e^{i\pi/3} \cdot e^{i\pi/6}$, (b) $\frac{8e^{i5\pi/6}}{2e^{i\pi/3}}$, (c) $(\sqrt{2}e^{i\pi/8})^4$.

Exercise 3.3 (**core — *a rotated square*). A square is centred at the origin with one vertex at $z_0 = 2 + i$. Find its other three vertices by repeatedly multiplying by i , and state the side length.

Exercise 3.4 (**core — *rotation preserves geometry*). Let $\alpha \in \mathbb{R}$. Show that for all $z, w \in \mathbb{C}$, $|e^{i\alpha}z| = |z|$ and $|e^{i\alpha}z - e^{i\alpha}w| = |z - w|$. What does the second identity say geometrically?

De Moivre and powers (Skills 3–4)

Exercise 3.5 (**core — *more powers*). Evaluate (a) $(1 - i\sqrt{3})^9$, (b) $(2 + 2i)^5$, (c) $\left(\frac{1+i}{\sqrt{3}+i}\right)^{12}$.

Exercise 3.6 (**core — *De Moivre for negative n*). Assuming De Moivre for positive integers, prove it for negative integers $n = -m$ ($m \in \mathbb{N}$). [Hint: $(\cos \theta + i \sin \theta)^{-1} = \cos \theta - i \sin \theta$.]

Roots and the fundamental theorem (Skills 5–6)

Exercise 3.7 (*drill — *assorted roots*). Find all: (a) square roots of $2i$; (b) cube roots of -1 ; (c) fourth roots of -1 ; (d) fifth roots of 32 (leave in exponential form).

Exercise 3.8 (**core — *square root the algebraic way*). The argument of $5 - 12i$ is not a standard angle, so the exponential method is awkward. Instead find both square roots of $5 - 12i$ in Cartesian form by equating real and imaginary parts of $(a + ib)^2$, and verify by squaring.

Exercise 3.9 (★★ core — a non-real right-hand side). Solve $z^3 = 1 + i$, leaving the roots in exponential form.

Exercise 3.10 (★★ core — a sixth-degree equation). Solve $z^6 = 64$, giving all six roots in Cartesian form, and describe their configuration.

Exercise 3.11 (★★ core — data-science flavour: roots of unity). In an N -point discrete Fourier transform the sampling frequencies are the N th roots of unity. Take $N = 4$ and $\omega = e^{i2\pi/4}$. (a) List $\omega^0, \omega^1, \omega^2, \omega^3$. (b) Show that $\sum_{k=0}^3 \omega^k = 0$ and $\sum_{k=0}^3 \omega^{2k} = 0$, whereas $\sum_{k=0}^3 \omega^{0 \cdot k} = 4$. [These “orthogonality” sums are what make the DFT invertible.]

Exercise 3.12 (★★ core — data-science flavour: twiddle factors). The fast Fourier transform is built from the “twiddle factors” $W_N^k = e^{-i2\pi k/N}$. Take $N = 8$. (a) Write W_8^1 in Cartesian form. (b) Show that $W_8^{k+4} = -W_8^k$ for every integer k (the symmetry that halves the FFT’s work). (c) Give the modulus and argument of W_8^3 .

Exercise 3.13 (★★ core — FTA and real factors). (a) Verify that $z = 2i$ is a root of $z^2 + 4$, and write down the other. (b) Factor $z^4 + 4$ into two real quadratics. [Hint: find the four roots first.]

Exercise 3.14 (★★★ challenge — a root-of-unity product). Let $\omega = e^{i2\pi/5}$. Show that $(1 - \omega)(1 - \omega^2)(1 - \omega^3)(1 - \omega^4) = 5$. Generalise to n .

Exercise 3.15 (★★★ challenge — a conjugate equation). Find all $z \in \mathbb{C}$ with $z^2 = \bar{z}$.

Trigonometric identities (Skill 7)

Exercise 3.16 (★★ core — $\cos 4\theta$). Use De Moivre to show $\cos 4\theta = 8 \cos^4 \theta - 8 \cos^2 \theta + 1$.

Exercise 3.17 (★★ core — $\cos^5 \theta$). Use the inverse Euler formula to express $\cos^5 \theta$ as a sum of cosines of multiples of θ .

Exercise 3.18 (★★★ challenge — $\tan 3\theta$). Show that $\tan 3\theta = \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}$.

Exercise 3.19 (★★★ challenge — a finite cosine sum). Using the geometric series for $\sum_{k=0}^{n-1} e^{ik\theta}$ (with $\theta \neq 0 \pmod{2\pi}$), show that

$$\sum_{k=0}^{n-1} \cos k\theta = \frac{\sin(n\theta/2)}{\sin(\theta/2)} \cos\left(\frac{(n-1)\theta}{2}\right).$$

Trigonometric–hyperbolic link (Skill 8)

Exercise 3.20 (★★ core — the converse pair). Show that $\cosh(ix) = \cos x$ and $\sinh(ix) = i \sin x$, and evaluate $\sin(2i)$ and $\cos(\pi i)$ in terms of e .

Exercise 3.21 (★★ core — identity transfer). Starting from $\cos 2\theta = 1 - 2 \sin^2 \theta$, set $\theta = ix$ and use the link of the previous exercise to derive the hyperbolic identity $\cosh 2x = 1 + 2 \sinh^2 x$.

Phasors (Skill 9)

Exercise 3.22 (★★ core — a 5–12–13 combination). Combine $5 \cos \omega t + 12 \sin \omega t$ into $R \cos(\omega t - \alpha)$, giving R exactly and α to three decimals.

Exercise 3.23 (★★ core — physics word problem: two AC sources). Two alternating voltages of equal amplitude but different phase, $V_1 = 10 \cos \omega t$ and $V_2 = 10 \cos(\omega t + \frac{2\pi}{3})$ (volts), are added. Find the amplitude and phase of $V_1 + V_2$.

Exercise 3.24 (**** challenge — balanced three-phase*). Show that three equal phasors 120° apart sum to zero: $1 + e^{i2\pi/3} + e^{i4\pi/3} = 0$. Interpret the result physically.

Concept checks (true/false and spot-the-error)

Exercise 3.25 (** drill — true or false*). Decide and justify briefly: (a) $|e^{i\theta}| = 1$ for every real θ . (b) $e^{i\theta} = e^{i\varphi}$ implies $\theta = \varphi$. (c) Every complex number has exactly three cube roots. (d) The n th roots of unity always include 1.

Exercise 3.26 (*** core — spot the error*). A student writes: “Since $(\cos \theta + i \sin \theta)^{1/2} = \cos \frac{\theta}{2} + i \sin \frac{\theta}{2}$ by De Moivre, the square root of $e^{i\theta}$ is $e^{i\theta/2}$.” What is wrong, and what is the correct statement?